28 V to 12 V / 60 W
BUCK DC/DC CONVERTER

Modeling, Closed-Loop Control, Protection Logic, Device-Level Losses and Cross-Validation

Platform	Scope
MATLAB/Simulink R2023a; Simscape Electrical	Switching power stage, PI control and protection
LTspice	Device stress, loss analysis and open-loop comparison
Methods	State-space averaging, Bode, pole-zero, transient and tolerance sweeps

TECHNICAL PROJECT REPORT

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1. Introduction and Objectives

This project develops and evaluates a non-isolated asynchronous buck converter stepping a nominal 28 V DC bus down to a regulated 12 V output at 60 W. The study progresses from sizing and open-loop characterization through PI voltage regulation, semiconductor-level loss evaluation, UVLO/OVP/OCP protection, frequency-domain control theory, component tolerance sweeps and LTspice–Simscape comparisons.

The objective is engineering comprehension and simulation-based assessment, not hardware qualification or a claim of MIL-STD-704F certification. Measured data and simplifying model assumptions are differentiated throughout.

2. Converter Architecture and Design Parameters

Parameter	Project value	Rationale
Nominal input / output	28 V / 12 V	DC bus conversion
Nominal output power / current	60 W / 5 A	12 V × 5 A
Load	2.4 Ω	12^2 / 60
Switching frequency	100 kHz	10 µs switching period
Inductor	47 µH	approximately 30% full-load current-ripple target
Capacitor	220 µF	output ripple and transient headroom
Normal reference input range	22–29 V	MIL-STD-704F-referenced test range
Power devices	High-side N-MOSFET + Schottky diode	asynchronous buck
Protection limits	UVLO 20 V; OVP 13.2 V; peak OCP 6 A	project-selected thresholds

2.1 Initial duty and component choice

$$D_{ideal} = V_{out} / V_{in} = 12 / 28 = 0.4286$$

$$I_{out} = P_{out} / V_{out} = 60 / 12 = 5\ A \quad ; \quad R_{load} = V_{out}^2 / P_{out} = 2.4\ \Omega$$

$$\Delta I_{L,target} = 0.30 \times 5 = 1.5\ A_{pp} \quad ; \quad I_{L,peak,ideal} \approx 5.75\ A$$

$$C_{min} \approx \Delta I_L / (8\ f_s\ \Delta V_{out,max}) \approx 15.6\ \mu F\ (\text{at } 120\ mV_{pp}\ \text{target})$$

The larger 220 µF capacitor was an intentional simulation design choice for ripple margin and transient energy storage; it is not a universal optimum. Nonideal switching models required duty near 0.47 in the original Simscape calibration, unlike the ideal 0.4286 calculation.

3. Analytical Power-Stage Modeling

3.1 Average continuous-conduction equations

Using inductor current iL and capacitor/output voltage vo as states, the ideal continuous-conduction averaged model is:

$$L \cdot di_L/dt = d(t)\ V_{in} - v_o(t)$$

$$C \cdot dv_o/dt = i_L(t) - v_o(t)/R$$

For d = D + δd and a fixed input voltage, linearizing about the steady operating point gives the duty-to-output transfer function:

$$G_{vd}(s) = \delta V_o(s)/\delta D(s) = V_{in} / [LCs^2 + (L/R)s + 1]$$

$$G_{vd}(s) = 28 / [1.034\times10^{-8}\ s^2 + 1.95833\times10^{-5}\ s + 1]$$

3.2 Natural dynamics

Metric	Value
Natural frequency	ω_n = 9834.22 rad/s; f_n = 1565.16 Hz
Damping ratio	ζ = 0.0963
Ideal poles	$-946.97 \pm j9788.52\ s^{-1}$
Ideal 1% duty-step output	0.28 V final change
Ideal duty-step overshoot	73.79%
Approx. 2% settling time	4.224 ms

The negative real pole parts represent a stable but lightly damped ideal plant. The normalized step response of Gvd is not a physical 0–12 V startup and cannot be used as such.

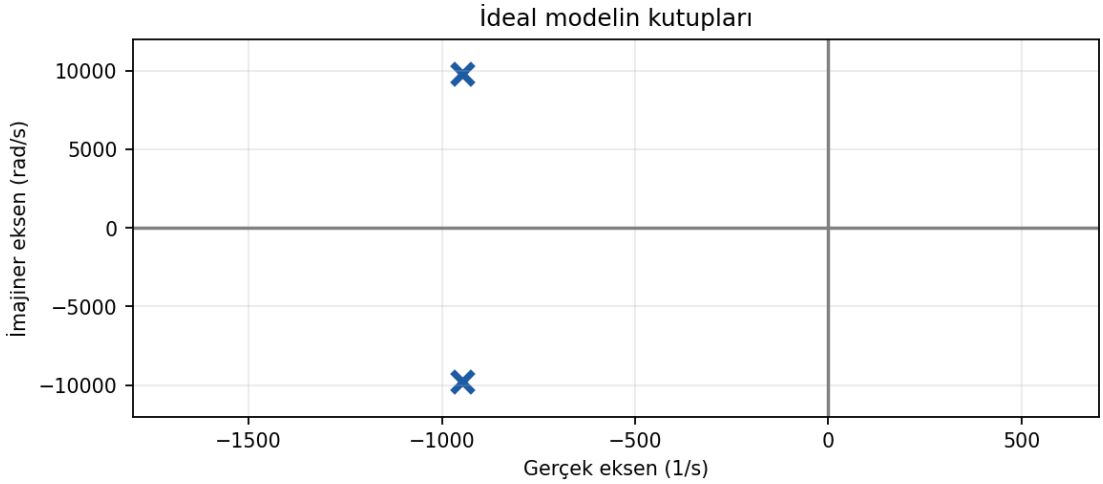


Figure 1. Ideal CCM duty-to-output plant pole locations.

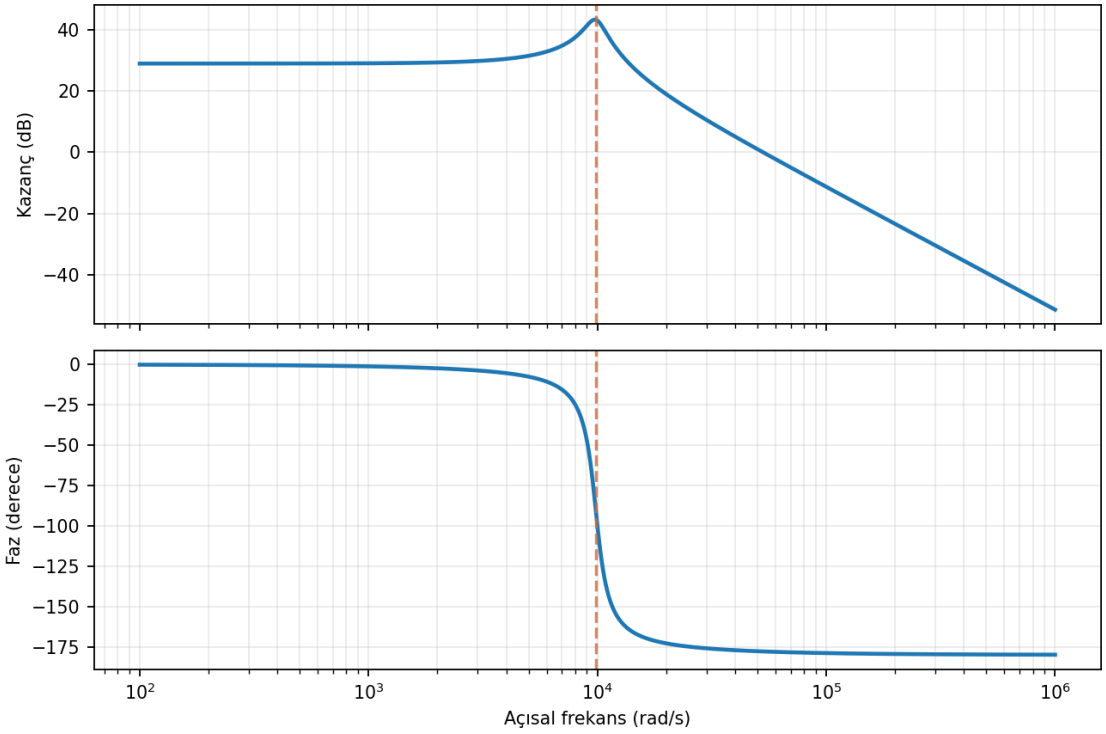


Figure 2. Ideal Buck plant frequency response showing LC resonance and phase transition.

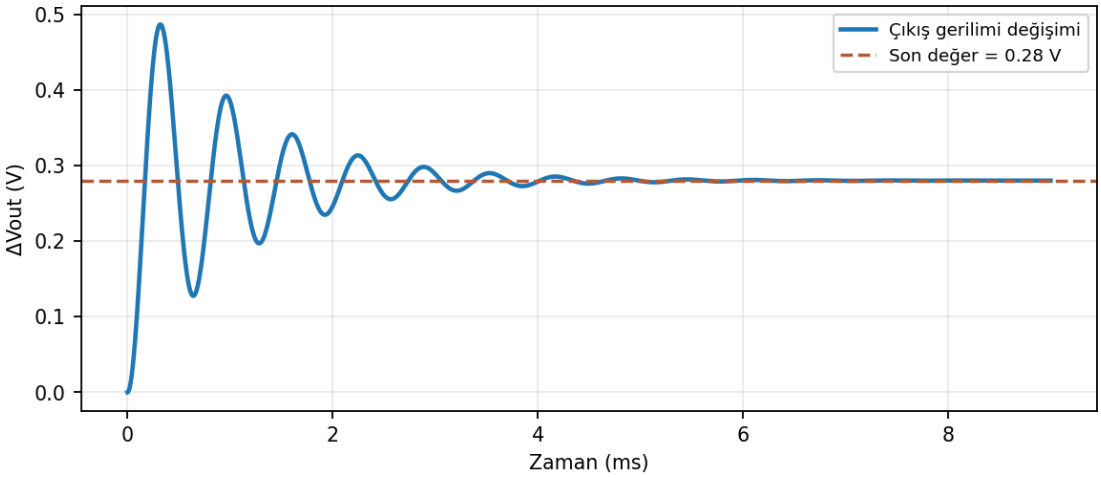


Figure 3. Small-signal duty-step response (Δd = 0.01); strong overshoot reflects low damping.

4. MATLAB/Simulink Implementation and Closed-Loop Regulation

4.1 Open-loop characterization

At fixed duty of approximately 47.3%, the output followed the applied input and load. This provided the baseline demonstrating why duty feedback is necessary.

Vin	Steady-state Vout	Error relative to 12 V
22 V	9.449 V	−21.258%
28 V	12.107 V	+0.889%
29 V	12.550 V	+4.581%

At nominal 28 V input, load variation from 25% to 100% produced steady-state output from 12.713 V down to 12.107 V, a spread of 0.606 V.

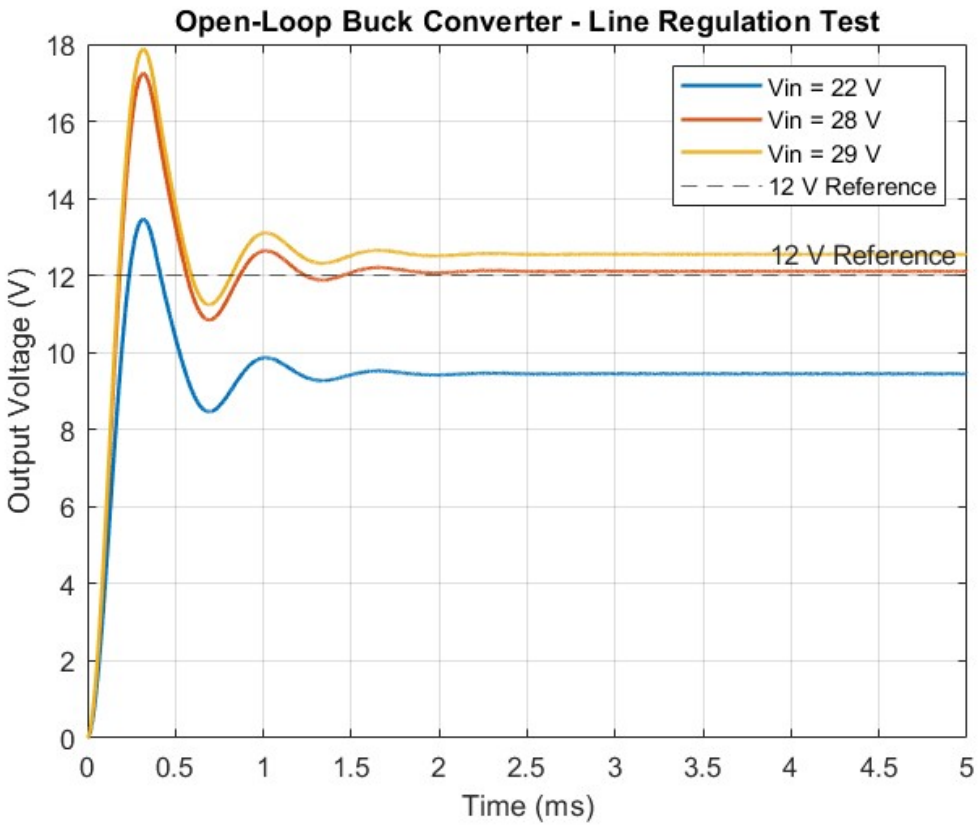


Figure 4. Open-loop line regulation: fixed duty cannot maintain a constant 12 V across 22–29 V.

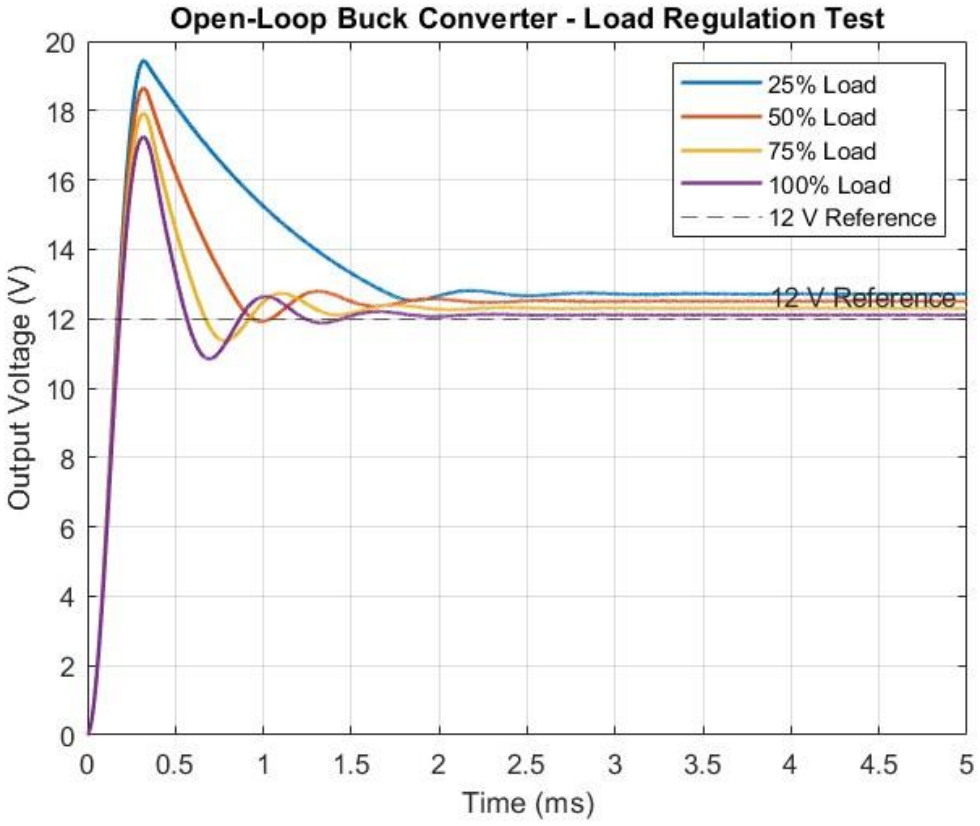


Figure 5. Open-loop load regulation across 25–100% load.

4.2 Feedback and duty command

$$e(t) = Vref(t) - Vout(t)$$

$$dcommand(t) = Kp\ e(t) + Ki \int e(t)dt \quad ; \quad 0 \le d \le 0.8$$

The implementation uses a voltage feedback comparator, PI control, duty saturation and a 100 kHz sawtooth comparator to produce PWM. Initial iterative tuning progressed from Kp = 0.02, Ki = 20 to Kp = 0.02, Ki = 50. These values were exploratory, not mathematically optimized.

PWM timing was checked with the period between like edges: approximately 9.998 µs or 100.015 kHz. Signal logging was used to export Vout and IL for time-response and steady-state calculations.

5. LTspice Device-Level Validation, Losses and Thermal Assessment

5.1 Device-level model and steady-state results

A high-side 100 V N-MOSFET (BSC100N10NSF class), MBR20100CT Schottky diode, inductor DCR and capacitor ESR/ESL were modeled in LTspice. The open-loop ON interval was iteratively refined near 4.494 µs for approximately 12 V at 100 kHz.

Measurement	Reported result
LTspice Vout after tuning	≈ 11.99–12.009 V (different tuning points)
Vout ripple	≈ 20–21 mVpp
Inductor current ripple	≈ 1.49 App
MOSFET VDS OFF	≈ 28.658 V
MOSFET conduction loss	≈ 0.114 W
MOSFET total modeled loss	≈ 0.154 W
Schottky loss	≈ 1.763 W

Switching-node voltage below ground during diode freewheeling explains why the MOSFET OFF-state VDS slightly exceeds the 28 V nominal input. Power loss estimates represent the device models and selected averaging windows, not final hardware dissipation.

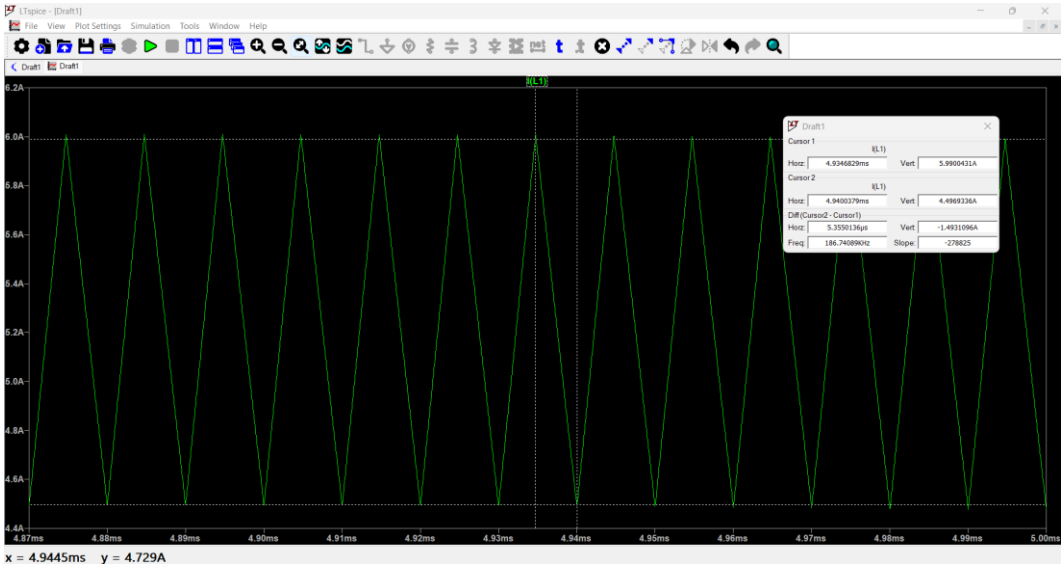


Figure 6. LTspice initial inductor current and output ripple during the tuning stage.

5.2 Loss breakdown and efficiency

$$P_{inductor,Cu} = I_{L,RMS}^2\ R_{DCR} = 5.0237^2 \times 0.0409 \approx 1.032\ W$$

$$\eta = P_{out} / P_{in} \times 100\%$$

Case	Pin	Pout	Simulated efficiency
Original generic 56 mΩ inductor	63.239 W	59.909 W	94.73%

40.9 mΩ DCR candidate	63.059 W	60.092 W	95.29%

Lower DCR reduced calculated copper loss from approximately 1.409 W to 1.032 W and correspondingly improved simulated efficiency. The loss balance also identifies the Schottky diode as a leading dissipative element.

5.3 Thermal interpretation

$$\Delta T = P_{loss} \times R\theta$$

At 1.763 W diode loss and the worksheet illustrative RθJA = 60 °C/W, ambient-to-junction rise is approximately 105.8 °C. Thus TJ ≈ 130.8 °C at 25 °C ambient and ≈ 190.8 °C at 85 °C ambient. These are thermal warnings based on assumed mounting conditions, not verified hardware temperatures. Device case, PCB copper, airflow and actual thermal ratings require assessment for implementation.

6. Protection Logic and Abnormal-Condition Tests

Gate = PWMraw AND UVLO_OK AND OVP_OK AND OCP_OK

Protection	Design value	Observed / intended behavior
Soft-start	12 V reference ramp over 2 ms	Peak IL 5.798 A; Vout 11.993 V steady state
UVLO	20 V input threshold	19 V input: output ≈0 V; 28 V: ≈11.996 V
Latched OVP	13.2 V output threshold	Trip 4.010 ms at 13.210 V in isolated 4.8 Ω test
Cycle-by-cycle OCP	6 A inductor peak	1.2 Ω overload: peak 6.088 A; Vout 6.489 V

The first simple comparator-based OCP allowed gate re-entry within one switching cycle and produced undesirable peak-current spikes. An S–R latch reset at each PWM rising edge prevents re-enabling the gate until the next cycle. A 0.1 µs unit delay addressed an algebraic-loop issue, with approximately 1.47% peak overshoot above the nominal 6 A threshold.

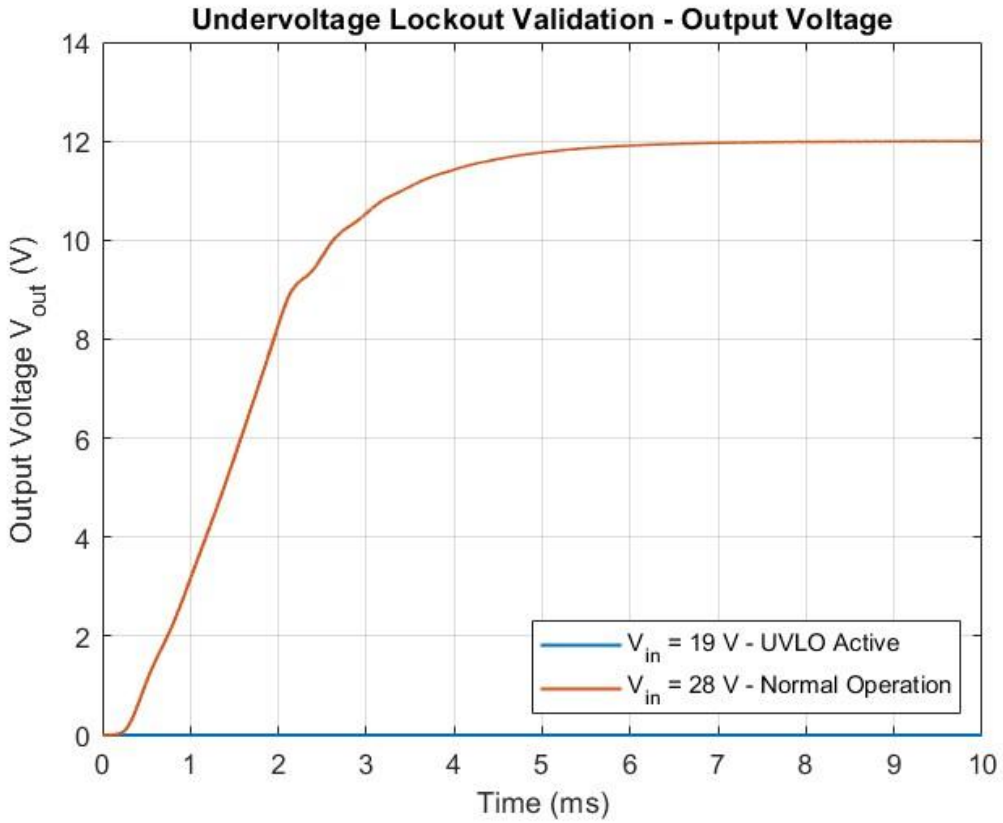


Figure 7. UVLO comparison showing shutdown below threshold and nominal regulation above threshold.

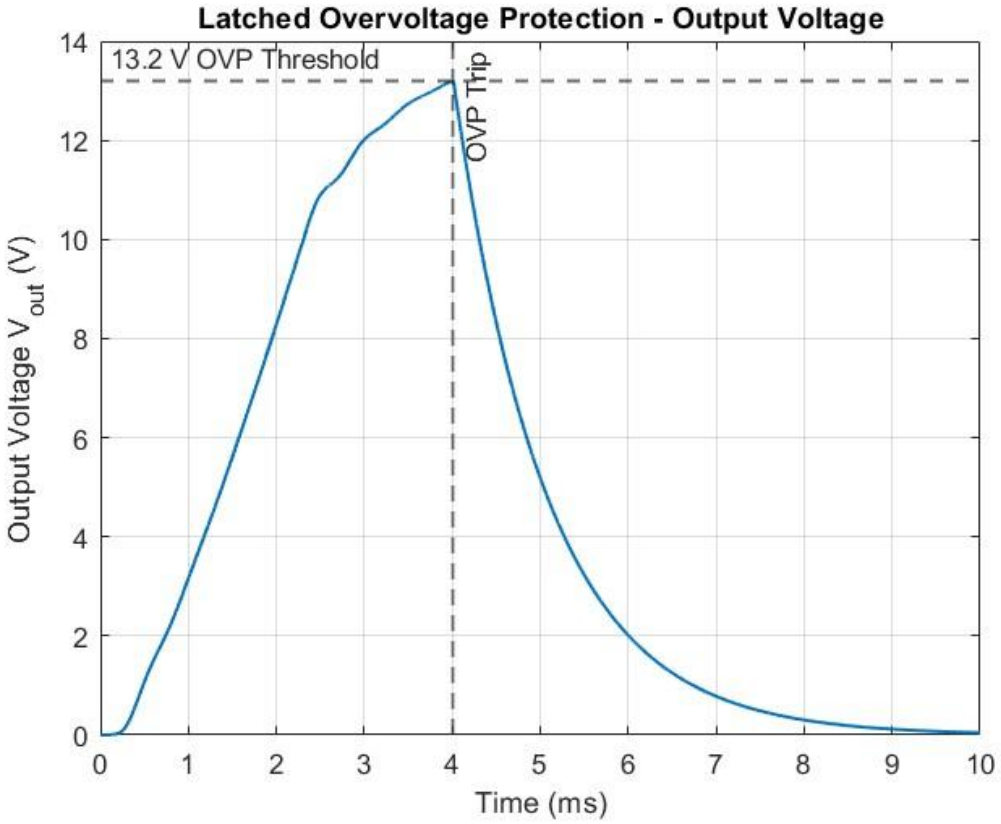


Figure 8. Latched OVP: voltage reaches trip point then output capacitor discharges after PWM shutdown.

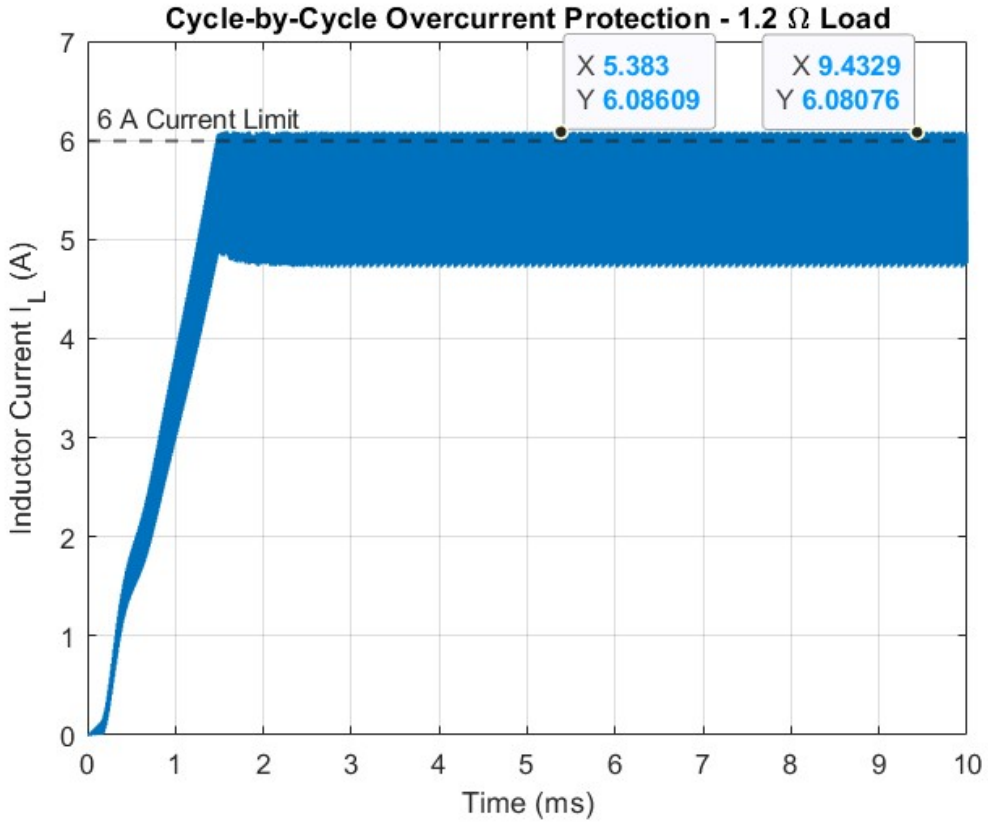


Figure 9. Cycle-by-cycle OCP full transient under 1.2 Ω overload.

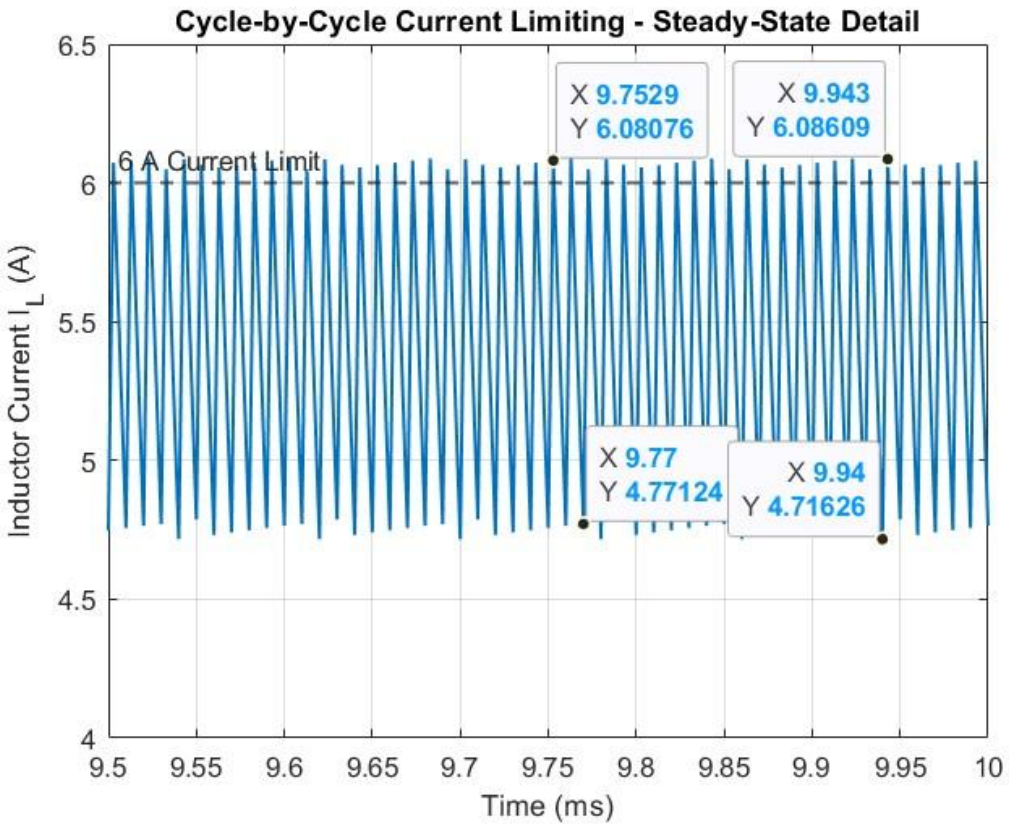


Figure 10. OCP switching-cycle detail, showing current limiting around the 6 A threshold.

7. Normal Input, Load and Referenced Transient Validation

7.1 Normal input variation

Vin	Vout average	Vout ripple	IL ripple	IL peak
22 V	11.9815 V	36.8 mVpp	1.2493 App	5.6200 A
28 V	11.9933 V	21.5 mVpp	1.5951 App	5.7979 A
29 V	11.9942 V	20.1 mVpp	1.6401 App	5.8210 A

Across the tested 22–29 V normal input range, the simulated closed-loop system held the average output near 12 V. The highest inductor ripple among these three cases was 1.6401 App at 29 V, about 32.8% of full-load current.

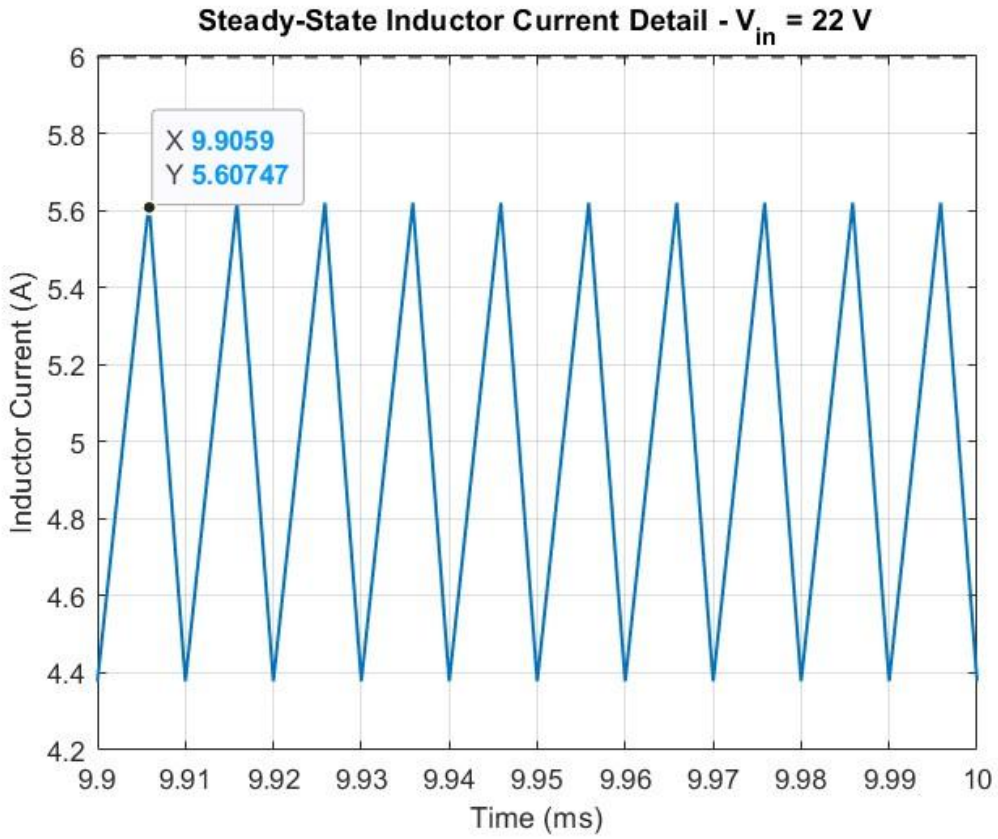


Figure 11. Steady-state inductor current at 22 V input (approximately 1.249 App).

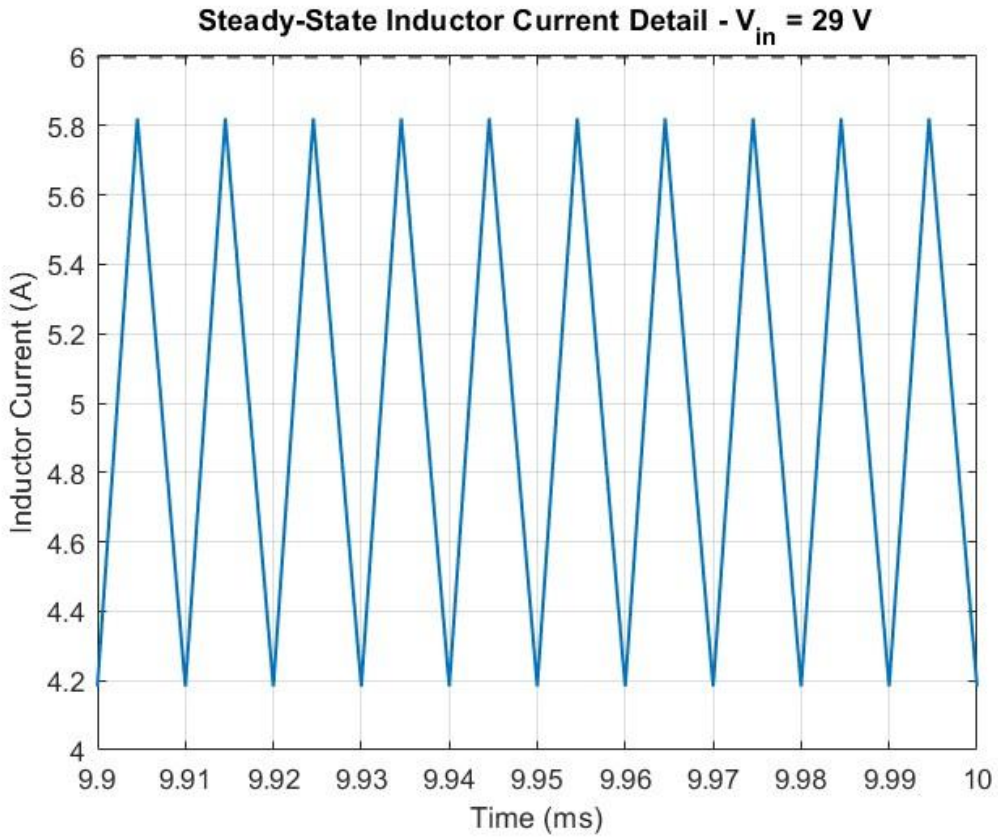


Figure 12. 29 V inductor current waveform at the tested ripple worst case.

7.2 Load-range regulation

Load	Resistance	Vout average	IL peak
25%	9.6 Ω	11.9936 V	2.0155 A
50%	4.8 Ω	11.9935 V	3.2760 A
75%	3.2 Ω	11.9934 V	4.5372 A
100%	2.4 Ω	11.9933 V	5.7979 A

The approximate 0.3 mV average-output difference across the four tested static loads is a result of closed-loop compensation. At the 25% load point, the estimated minimum inductor current remained positive (~0.49 A), consistent with CCM in this simulation.

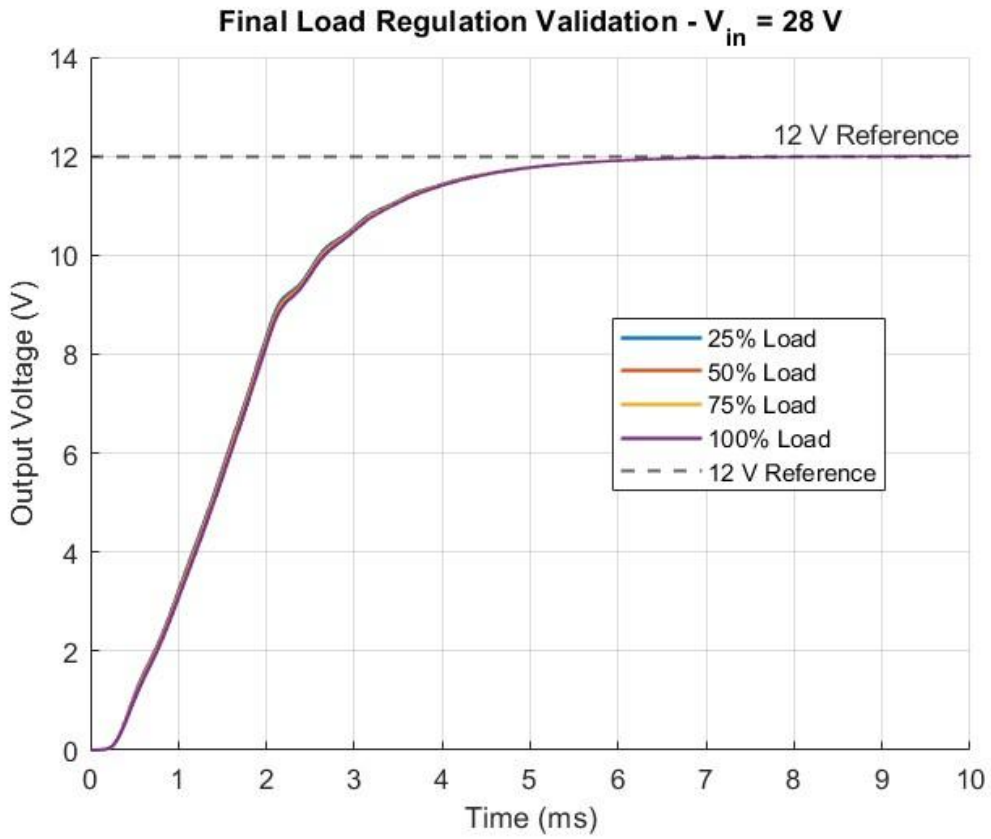


Figure 13. Closed-loop output responses at the four static load levels.

7.3 MIL-STD-704F-referenced undervoltage transient

The worksheet applies a deliberately simplified sequence: 28 V for 0–5 ms, 18 V for 5–20 ms, then 28 V for 20–40 ms. Since the selected project UVLO threshold is 20 V, the converter intentionally shuts down during the 18 V interval. This experiment is a standard-referenced characteristic test, not full MIL-STD-704F compliance.

Initially, UVLO only blocked the gate. The PI integrator continued accumulating error and its duty command reached the 0.8 saturation bound. Applying ePI = (12 − Vout) × UVLO_OK held the integrator error at zero during shutdown; the saved integral state remained. On restart, proportional action drove a temporary duty increase and approximately 12.59 V output peak (~4.9% overshoot). The output later returned to approximately 12.004 V.

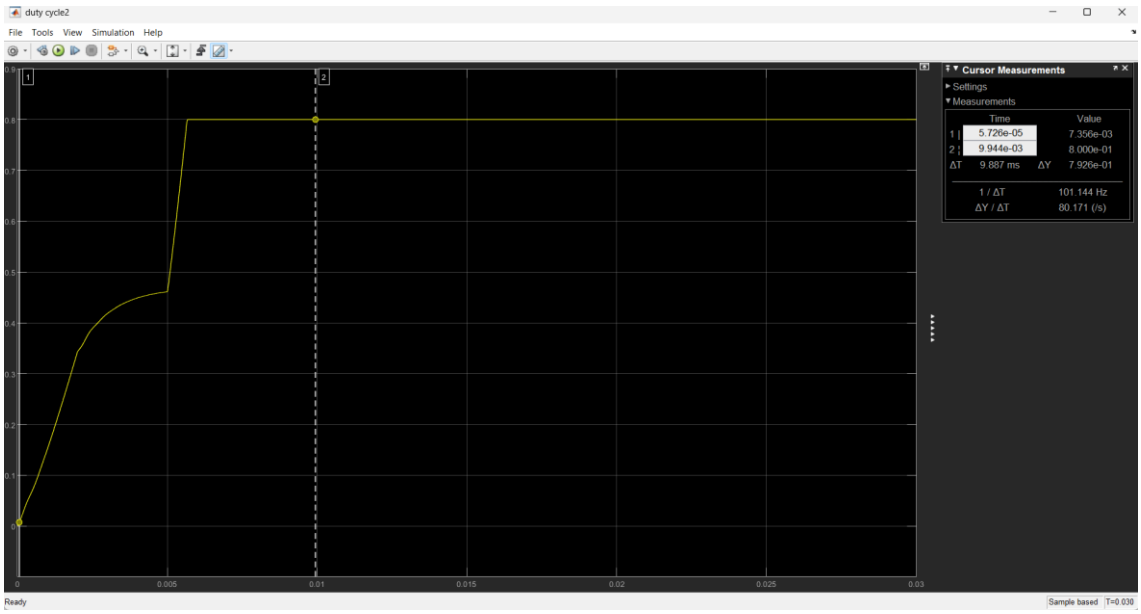


Figure 14. Duty saturation during UVLO before PI error gating.

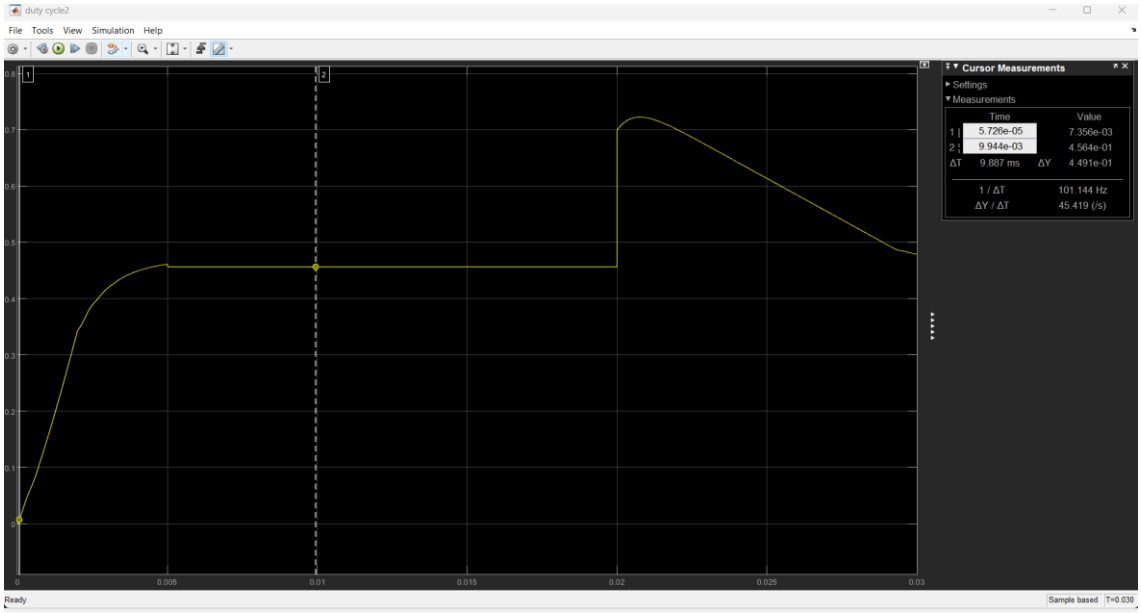


Figure 15. Duty after error gating: integrator accumulation is suppressed during UVLO.

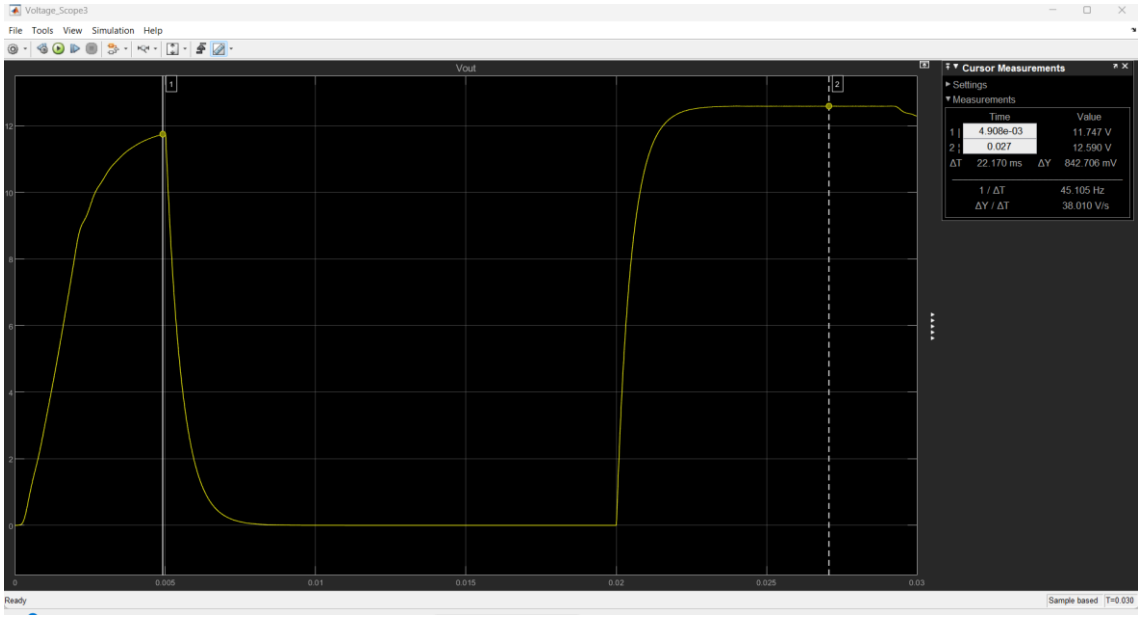


Figure 16. Restart output shows finite transient overshoot despite UVLO error gating.

8. Level 5 Control Theory and PI Design

8.1 Nonideal averaged plant

To isolate parasitic effects, an averaged plant incorporating L DCR (40.9 mΩ), capacitor ESR (12.2 mΩ), MOSFET resistance (10 mΩ) and an approximate diode drop was derived. The simplified nonideal transfer function in the Level 5 worksheet is:

$$G_{vd,nonideal}(s) \approx (7388s + 2.753\times10^{9}) / (s^{2} + 3114s + 9.805\times10^{7})$$

These numerical coefficients belong to an approximate averaged model; the switching Simscape diode subsequently showed a different parameterization (Vf = 0.6 V, Ron = 0.3 Ω). A difference in predicted DC operating points was expressly retained as a model limitation.

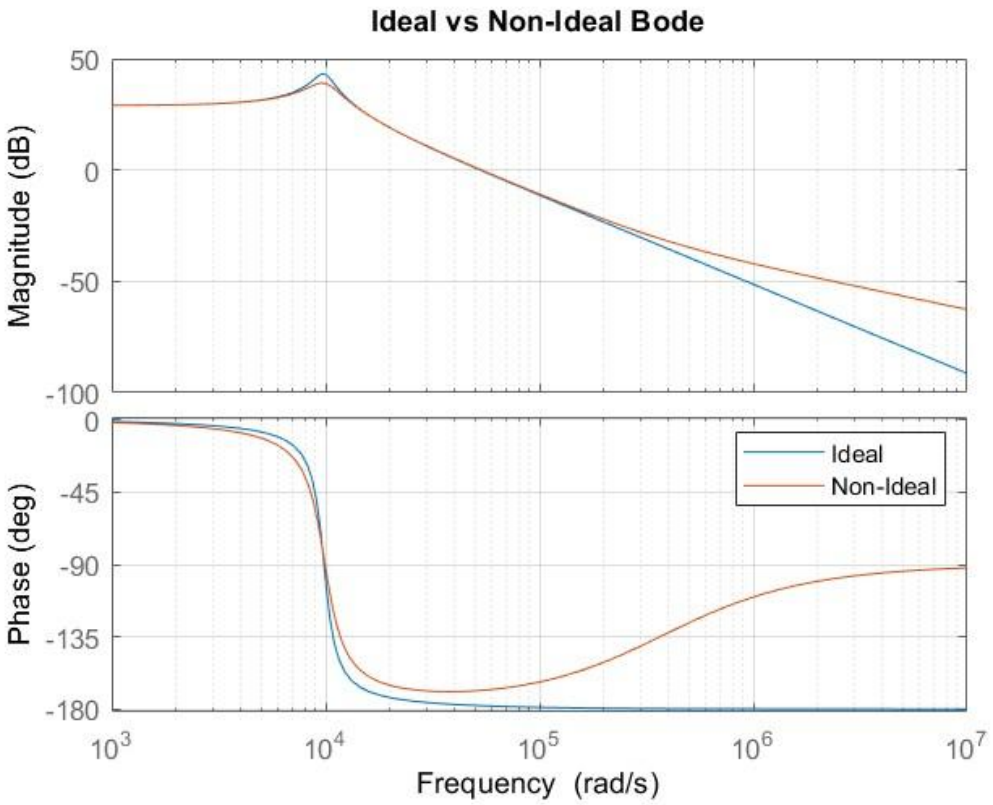


Figure 17. Bode comparison of ideal and simplified nonideal averaged Buck plants.

8.2 Original PI stability margins

Original PI metric	Value
Kp / Ki	0.02 / 50
Minimum phase margin	30.89°
Gain crossover	1886.88 Hz
Gain margin	∞ dB in analyzed loop model
Linear closed-loop settling	3.755 ms
Linear step overshoot	0%

The loop crossed unity gain above the approximately 1.55–1.57 kHz LC resonance region, where the plant contributed substantial phase lag. The PI contributed approximately 11.9° additional lag at crossover. A positive 30.89° margin does not imply instability, but motivates a more robust design goal of roughly 55–65°.

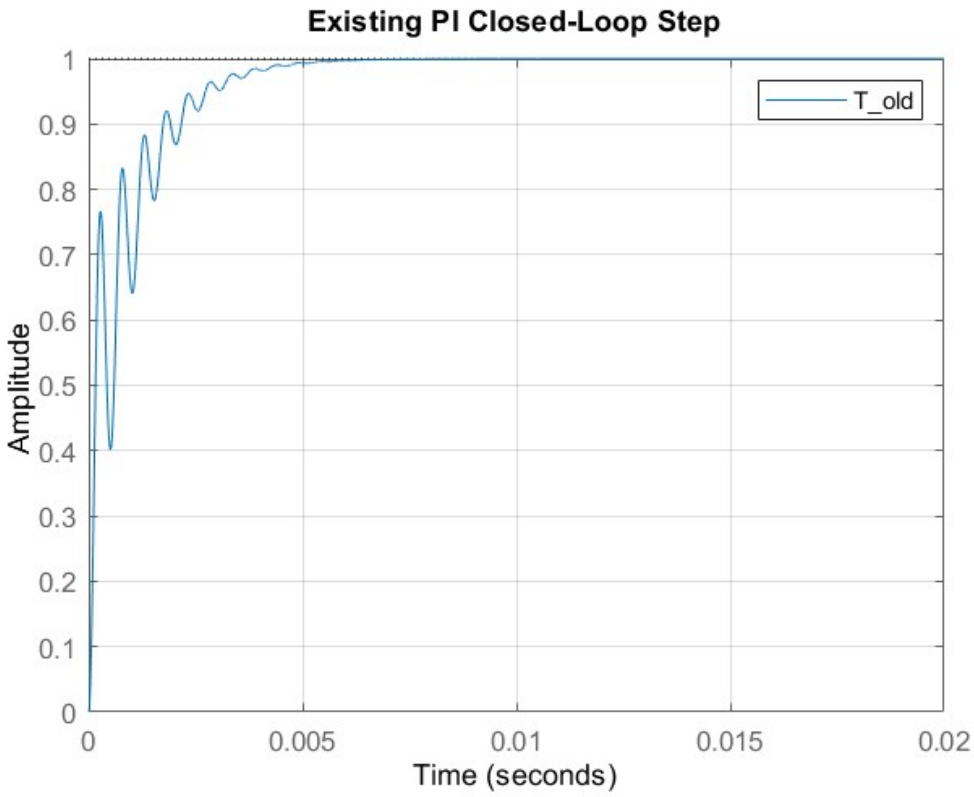


Figure 18. Original PI loop Bode and normalized response, showing low stability margin.

8.3 Rejected candidates and multi-crossover analysis

A 1300 Hz candidate (Kp ≈0.001993, Ki ≈118.596) achieved ~60.01° at one crossover, but produced additional crossings at 622.71 Hz (~85.99°) and 1625.97 Hz (~0.07°). Therefore the minimum margin—not the single target crossover—led to rejection. A separate 500 Hz candidate also suffered a critical high-frequency crossing (~28.57°).

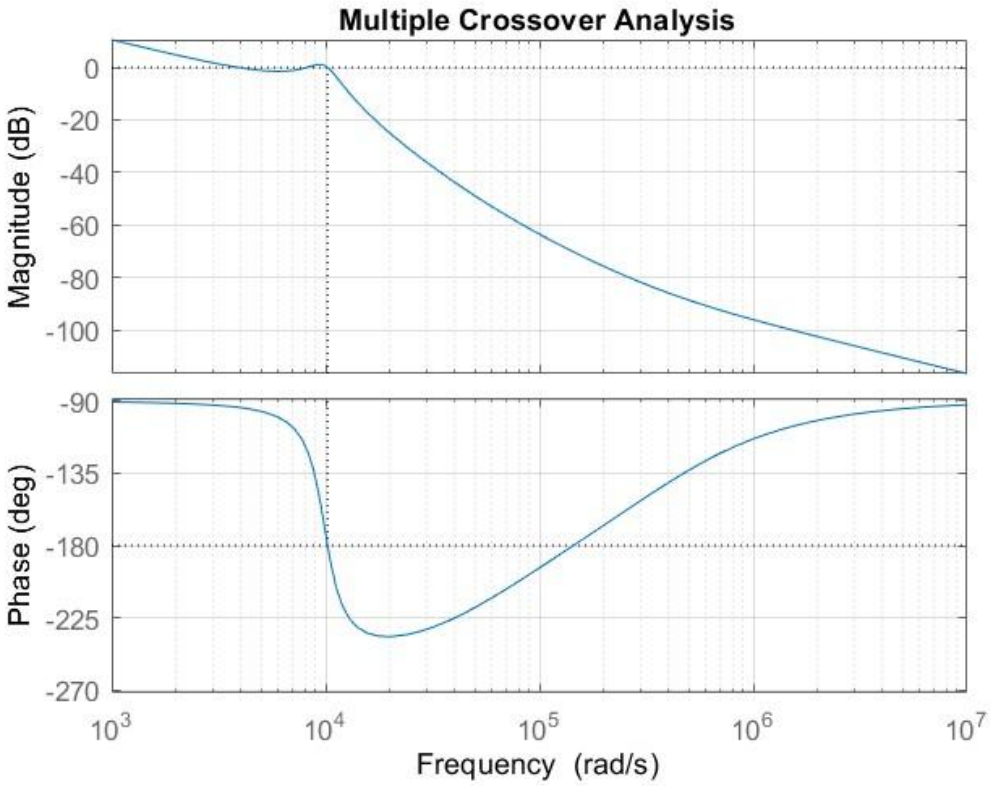


Figure 19. Illustration of multiple unity-gain crossings in a PI candidate.

8.4 Parameter sweep candidates

Controller	Kp	Ki	Minimum PM	Linear Ts	Linear OS
Original	0.020	50	30.89°	3.755 ms	0%
Candidate A	0.003	55.8546	92.19°	2.708 ms	0.340%
Candidate B	0.010	55.8546	56.68°	3.007 ms	0.033%

The original worksheet reports practically identical Simscape startup metrics for A and B (about 12.0000 V steady output, 12.0044 V peak and 4.144 ms 2% settling). The cause was not conclusively resolved; any claim that A is universally superior would therefore be unsupported. Candidate A was carried forward as a provisional choice.

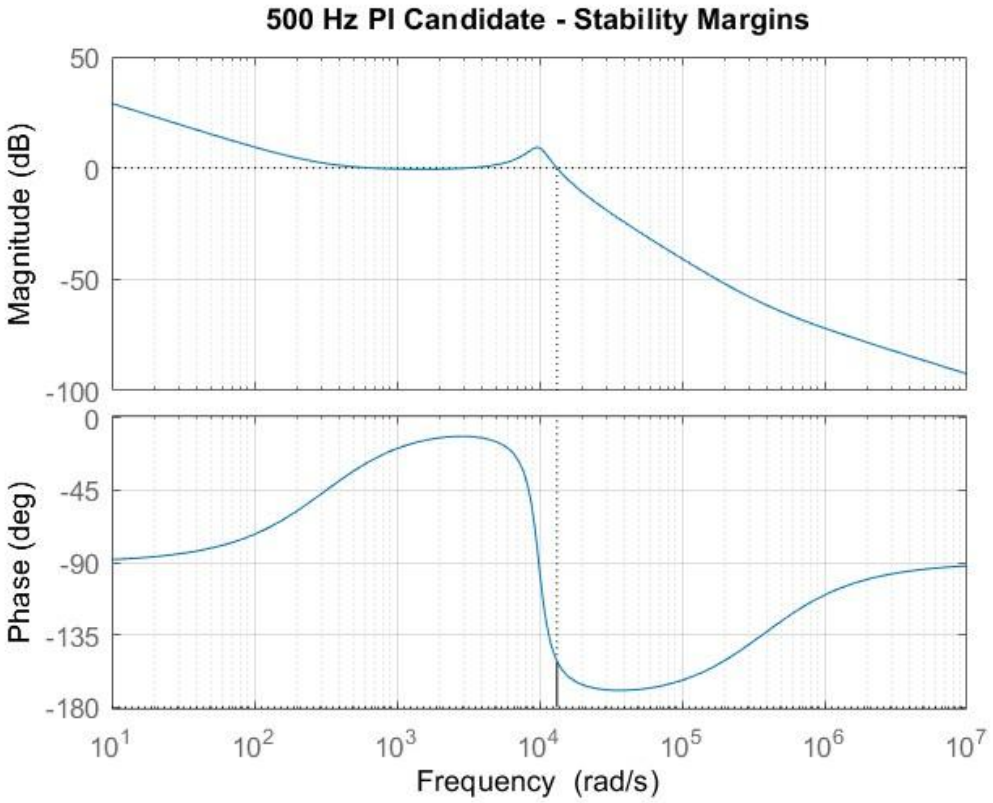


Figure 20. Original PI and candidate normalized linear closed-loop step responses.

9. Sensitivity, Tolerance and Cross-Platform Comparison

9.1 Line-step sensitivity

The later Level 5 line transient used 28 → 26 → 30 → 28 V at 20, 30 and 40 ms. The observed output ranged from 10.9219 V to 12.4842 V, with a maximum absolute deviation of 1.0781 V (8.984%) from 12 V. This is line-disturbance rejection, not startup overshoot.

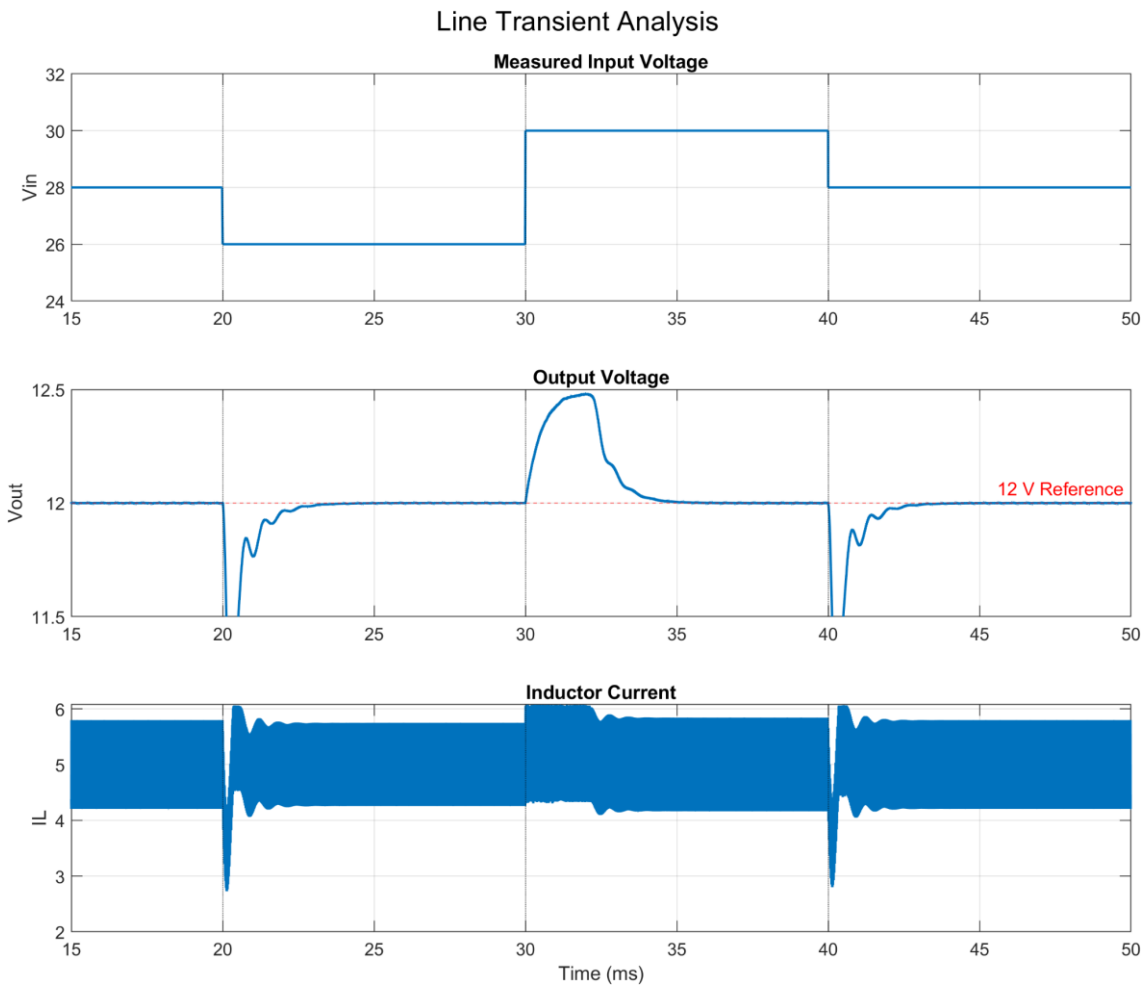


Figure 21. Level 5 line transient: applied input steps, output deviation and inductor current.

9.2 Component tolerance sweeps

Nine combinations of L = 37.6/47/56.4 µH and C = 176/220/264 µF were evaluated. The recorded overshoot range was approximately 0.0256–0.0577%, and 2% settling times about 3.7646–3.8339 ms for this sweep. An additional 3×3 sweep varied DCR between 32.72 and 49.08 mΩ and ESR between 9.76 and 14.64 mΩ. Startup responses nearly overlapped; the complete numerical DCR/ESR results were not included in the source worksheet.

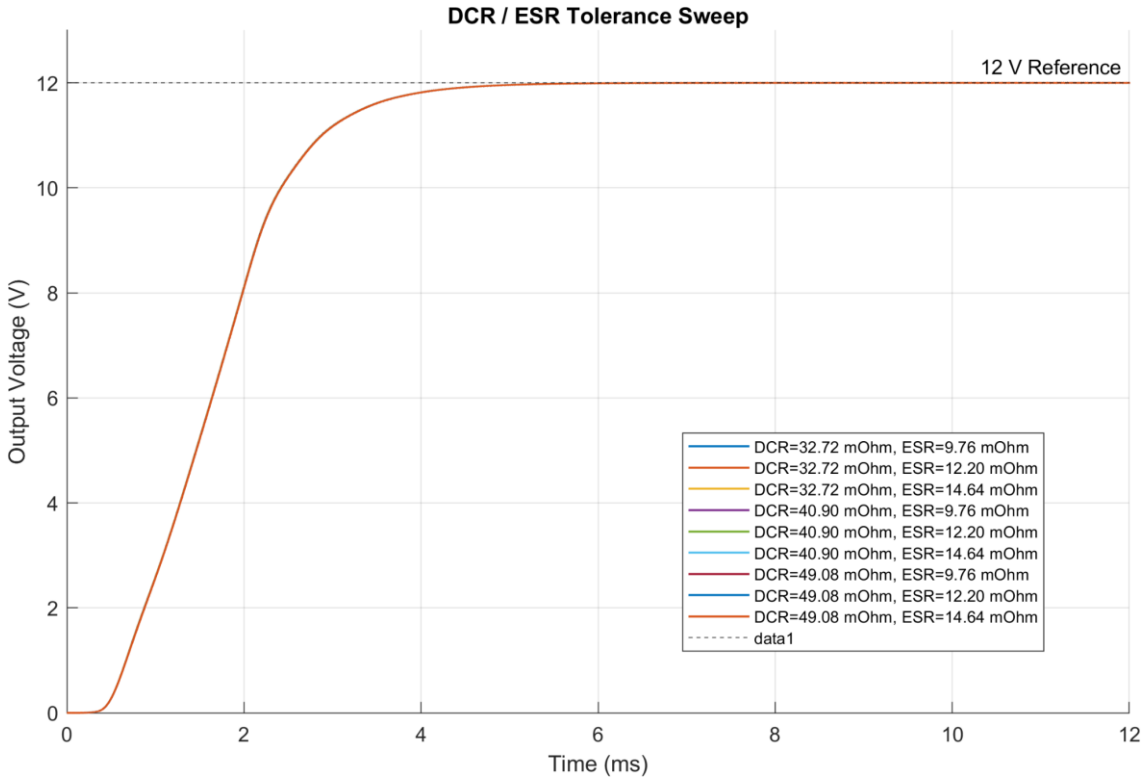


Figure 22. Nine-case L/C ±20% startup tolerance sweep.

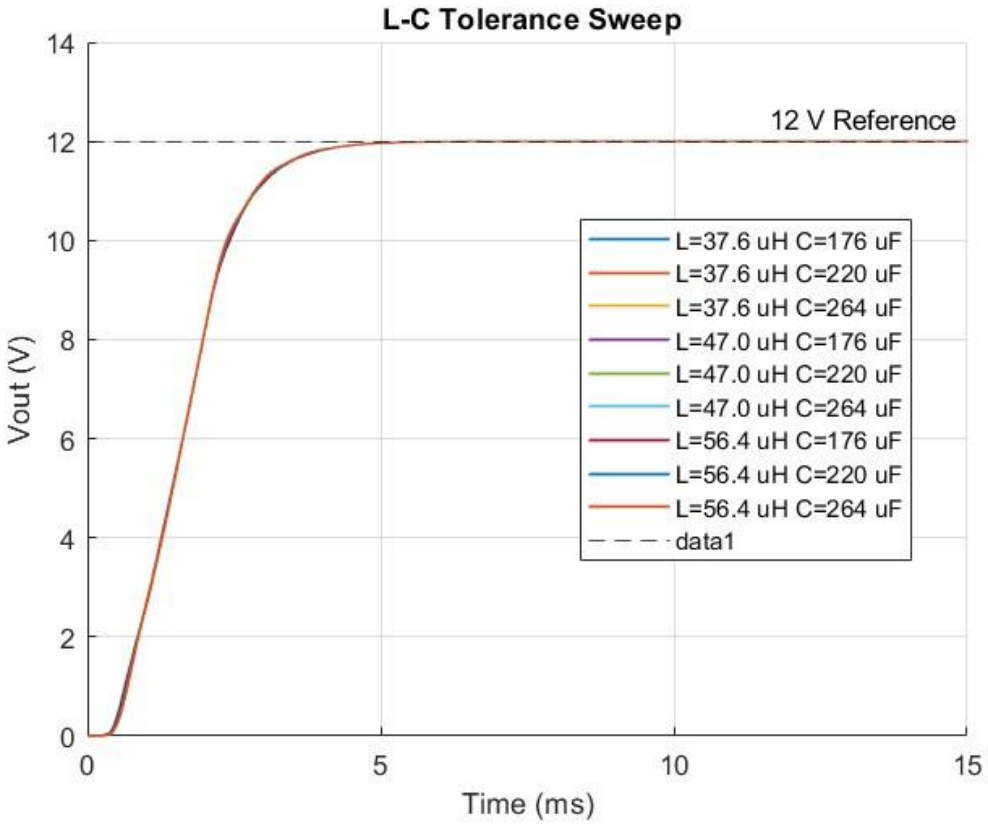


Figure 23. Nine-case DCR/ESR ±20% comparison.

9.3 LTspice–Simscape matched-duty comparison

For a meaningful comparison, both models were run open-loop with D = 0.4692, fs = 100 kHz, Vin = 28 V, Rload = 2.4 Ω and the same nominal L/C. Statistics were collected over the final 200 µs of the 5 ms run.

Metric	LTspice	Initial Simscape	Simscape diode test
Average Vout	12.629514 V	11.811520 V	12.552798 V
Vout ripple	22.6145 mVpp	19.420 mVpp	18.891 mVpp
Average IL	5.2634 A	4.9213 A	not remeasured
Switch-node minimum	−0.633 V	−2.3158 V	−0.6599 V

The initial average-output difference was ~6.48%. The Simscape piecewise-linear diode used Vf = 0.6 V and Ron = 0.3 Ω, producing an OFF-state drop much larger than the LTspice Schottky. Reducing Ron to 0.01 Ω as a sensitivity test moved the Simscape output to 12.552798 V, leaving ~0.61% DC mismatch. The replacement parameter does not establish full device-model equivalence; ripple differences remain.

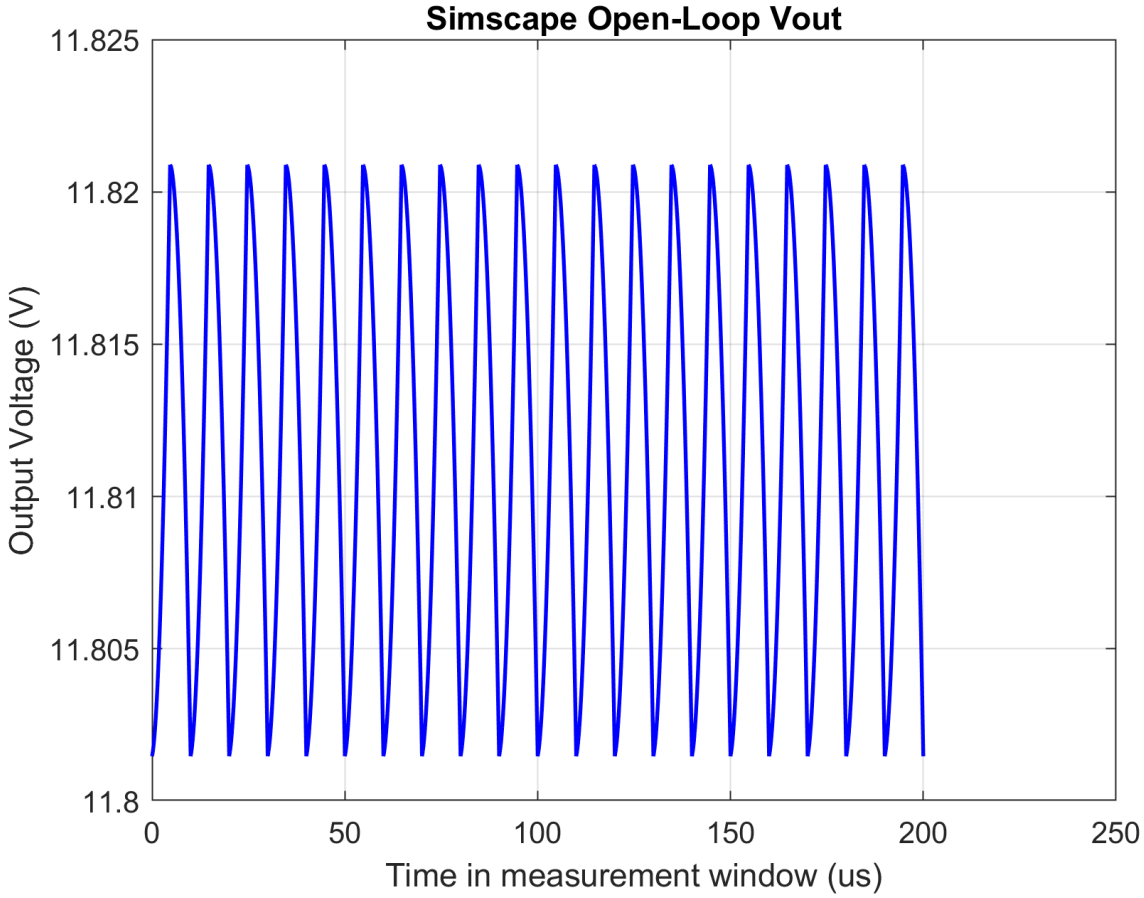


Figure 24. Initial Simscape open-loop output ripple at the common fixed duty.

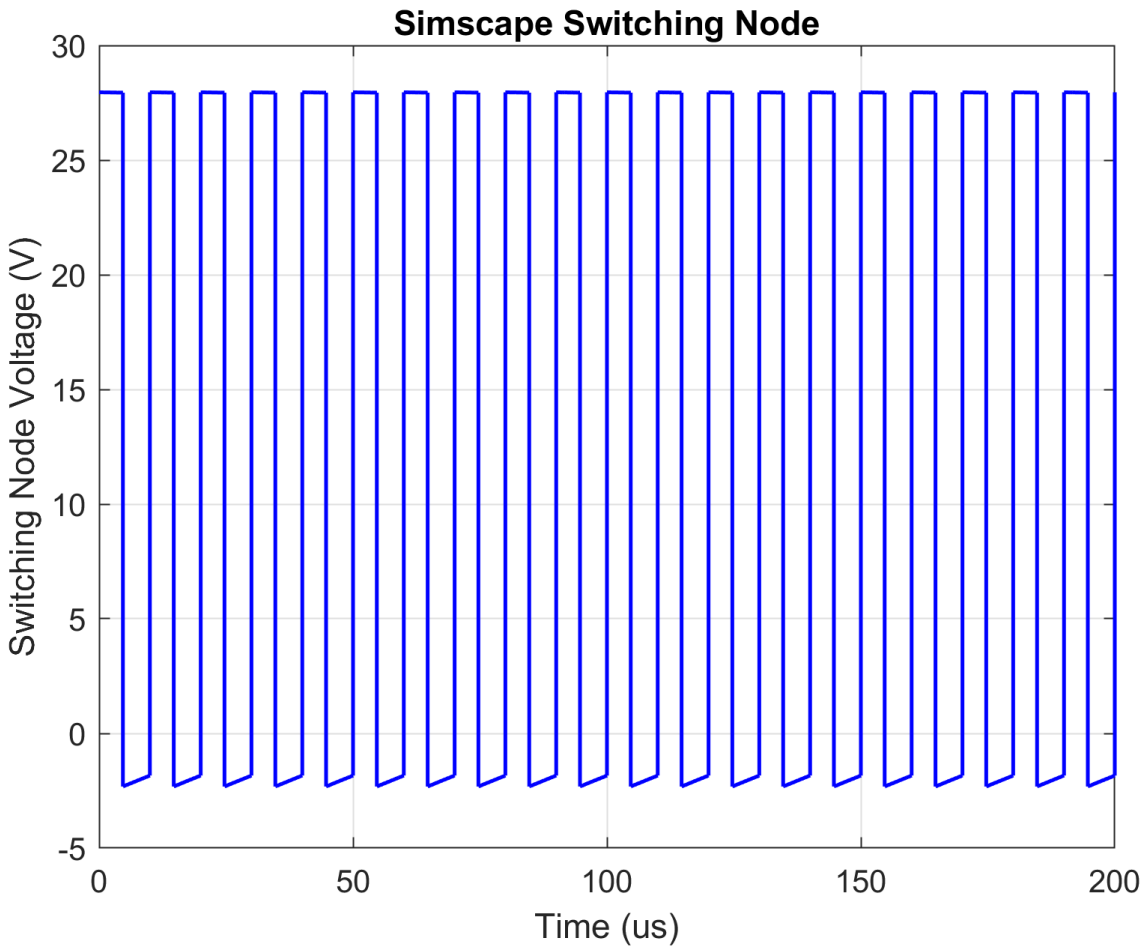


Figure 25. Simscape current ripple measured in the matched-duty comparison.

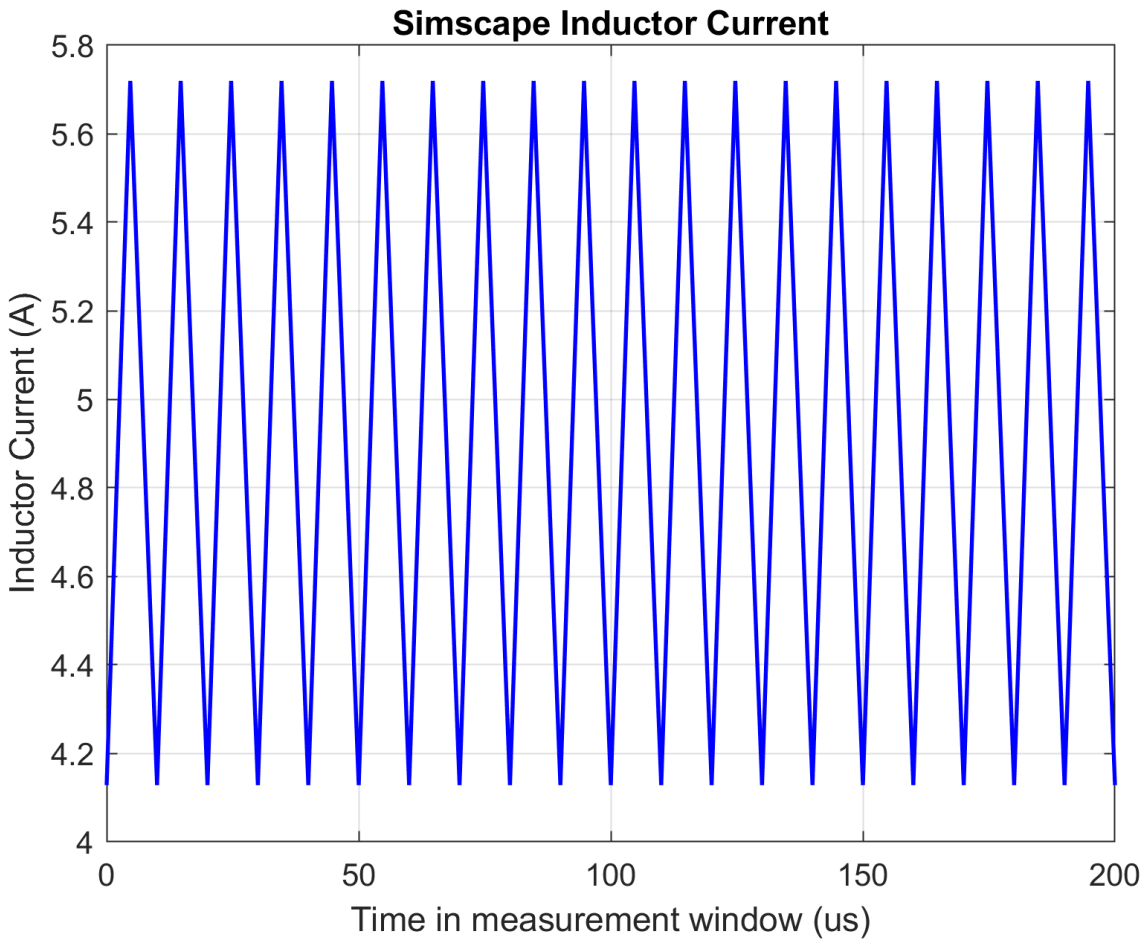


Figure 26. Simscape switch-node waveform showing the initial excessive negative diode conduction level.

10. Simulation Results and Consolidated Tables

The principal validated simulation outcomes are summarized below. Values are drawn from separate cases and should not be treated as one single matched model configuration.

Technical area	Main result	Scope
Open-loop behavior	9.449 to 12.550 V for 22–29 V	fixed duty
Closed-loop line regulation	11.9815–11.9942 V	22/28/29 V static inputs
Closed-loop load regulation	11.9933–11.9936 V	25–100% static load
Soft-start	5.798 A peak; ≈0.017% Vout overshoot	nominal Simscape
Protection	OVP at 13.210 V; OCP peak 6.088 A	isolated fault tests
Simulated efficiency	95.29%	LTspice, updated DCR candidate
Old PI minimum PM	30.89°	linearized nonideal model
Model alignment	6.48% → 0.61% DC error	single fixed-duty sensitivity test

11. Discussion, Limitations and Troubleshooting

11.1 Interpretation

The exercise shows why performance must be assessed at multiple modeling layers. Ideal CCM equations expose LC resonance, while switched Simscape simulations reveal startup, ripple and protection behavior. LTspice yields device waveform and approximate loss estimates. The fixed-duty comparison demonstrates that nominally identical converter parameters can still conceal materially different diode models.

11.2 Common issues and resolutions

Issue encountered	Root cause / explanation	Resolution used
Output varied with input or load	Fixed open-loop duty	Added PI voltage feedback
UVLO test ignored workspace Vin	DC source still set to fixed 28 V	Bound source parameter correctly
OVP failed to trip at full load	OCP limited first	Isolated OVP test with 4.8 Ω load
OCP chattering and current spikes	Comparator re-enabled gate within cycle	Cycle-by-cycle S–R latch
Algebraic-loop solver error	Zero-delay current-gate-plant feedback	0.1 µs Unit Delay
UVLO restart windup	PI integral accumulated while gate disabled	UVLO-based error gating
Line transient appeared as ramp	Workspace signal interpolation	Disabled interpolation and held final input value
LTspice / Simscape DC mismatch	Different diode conduction model	Compared switch-node and tested lower diode Ron

11.3 Evidence boundaries

No physical prototype or experimental oscilloscope data was presented. Source records do not provide complete efficiency curves at all input/load corners, complete DCR/ESR numeric tables, comprehensive MIL-STD-704F Figure 13/14 compliance matrices or a final validated matching of all semiconductor and gate-driver dynamics. The identical observed Simscape startup metrics for PI Candidates A and B remain unresolved. These boundaries are explicit rather than filled with inferred results.

12. Conclusion

A 28 V to 12 V / 60 W asynchronous Buck converter was modeled and characterized across circuit theory, nonlinear switched simulation, protective supervisory logic, frequency-domain controller design and LTspice semiconductor modeling. The tested Simscape model maintained near-12 V regulation at 22/28/29 V and four static load points; soft-start and protection functions demonstrated their selected shutdown/current-limit behavior.

The most informative engineering findings were the low initial 30.89° loop phase margin, dangerous multi-crossover behavior of one nominal 60° PI candidate, restart duty behavior caused by saved integral state, the efficiency effect of inductor DCR and the strong impact of diode Ron on LTspice–Simscape DC alignment. These results support a substantive simulation-based learning and analysis project, without implying full hardware or military-standard qualification.